

Sensor Fusion-Based Pedestrian Collision Warning System with Crosswalk Detection

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Abstract—This paper describes a pedestrian collision warning system with crosswalk detection feature based on sensor fusion of a monocular camera and a millimeter wave radar. The method to decide about the presence of a pedestrian is based on the assumption that objects moving along a crosswalk can be interpreted as pedestrians under certain circumstances. The advantage of the described solution is its robustness and effectiveness since it is limited to crosswalks. The camera can be used to detect the crosswalk. Data from both sensors can then be used to infer about the presence of a pedestrian. The sensor fusion algorithm which combines data from the sensors is explained. Then the paper describes a warning concept which provides auditory alarm and visual information about the presence of a crosswalk as well as pedestrians to a driver, depending on the estimated collision probability. Finally, test drives on an experimental vehicle are presented and the results verify that the proposed warning system is running in practice.

to measure the pedestrian position accurately. So a sensor fusion of camera and radar is a promising solution in order to get high reliability of the system.

In addition, the data of accidents in Japan reveals that 50% of pedestrian accidents have occurred near crosswalk, and most of them were in the situation that the vehicles were moving straight forward [2]. So this paper focuses on the situation when a vehicle approaches a crosswalk with a pedestrian on it. This approach has the advantage of high robustness, low false alarm rate and cost efficient implementation. The experimental vehicle detects the pedestrian near the crosswalk and estimates the collision probability, and then, supplies drivers with warnings in the case of danger situation. The rest of the paper is organized as follows. Section II gives the description of the test vehicle. Section III introduces the pedestrian collision warning system design. Section IV presents the experimental results. Finally, Section V provides some conclusions.

I. INTRODUCTION

According to statistical data of accidents, pedestrian fatalities are a particular priority in Europe and Japan, as shown in Figure 1 and Figure 2 [1-2]. Recently, researches on driving assistance systems which are method of technical solutions for accident prevention have been conducted extensively [3]. To detect pedestrians, existing researches make use of stereo camera, monocular camera, infrared camera and LIDARs as described in the survey by Gandhi and Trivedi [4]. Cameras have advantages of wide view range and high resolution power, so cameras can recognize attributes of objects by image processing. However there are some problems of image processing that, the algorithm is so complex, the computational cost is high, and the distance to target might be inaccurate. On the other hand, a LIDAR is not suitable to recognize attributes of objects, but it can measure the distance accurately. For development of pedestrian protection system in the situation of car-to-pedestrian accidents it needs calculation of the collision probability, therefore it is important for the system

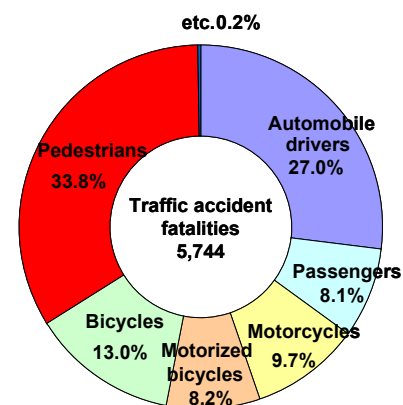


Fig.1 Statistical data of pedestrian fatalities in Japan [2]

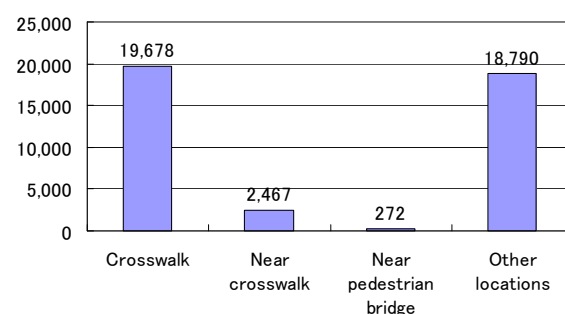


Fig.2 Location of pedestrian accidents (ITARDA 2007) [2]

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II. EXPERIMENTAL VEHICLE

The experimental vehicle used in this study was equipped with a millimeter wave radar, a monocular CMOS camera, PC, and a sensor cluster which gets the information of vehicle speed and yaw rate. The millimeter wave radar and the camera used in the study are Continental ARS300 and CSF200 respectively. The sensors and the measurement system configuration on the experimental vehicle are shown in Figure 3.

The 77GHz millimeter wave radar has two ranges. One is a long distance scanning range which has a measurement angle of 18 degree, and a detection range of 200 meters. The other is a short distance scanning range which has a measurement angle of 56 degree, and a detection range of 60 meters. On this sensor an algorithm for object detection and tracking is implemented. This algorithm segments feature points which are recognized by millimeter wave radar, and outputs the position data by tracking with Kalman filter. Thus, the position and the relative velocity of detected objects can be obtained by the millimeter wave radar. These data are input into a PC via a CAN-bus device and can be monitored in real time. The image from the monocular camera is captured by a measurement interface in real-time and the image is then delivered into the PC for the image processing task. The field of view of the camera is 42 deg (H) x 26 deg (V) with the resolution of 752 x 480 pixels, and its update rate is 40 milliseconds.

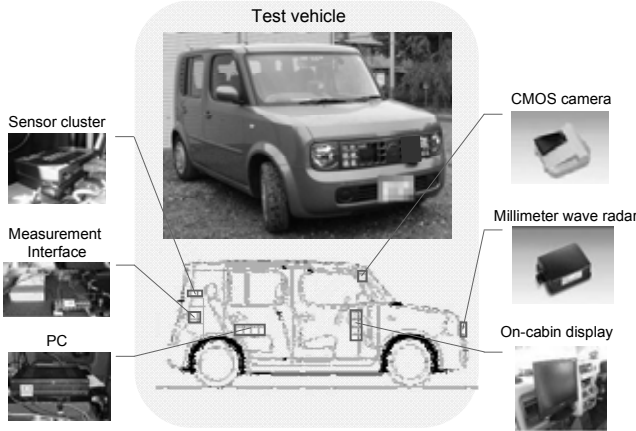


Fig.3 Experimental vehicle equipped with CMOS camera, millimeter wave radar, measurement system and display

III. PEDESTRIAN DETECTION AND WARNING SYSTEM DESIGN

The architecture of the pedestrian collision warning system is shown in Figure 4. This system consists of three algorithms: a crosswalk detection algorithm, a sensor fusion algorithm of camera and millimeter wave radar, and an algorithm for calculation of the probability of pedestrian collision. A warning is provided to the driver when the pedestrian collision reaches a certain level.

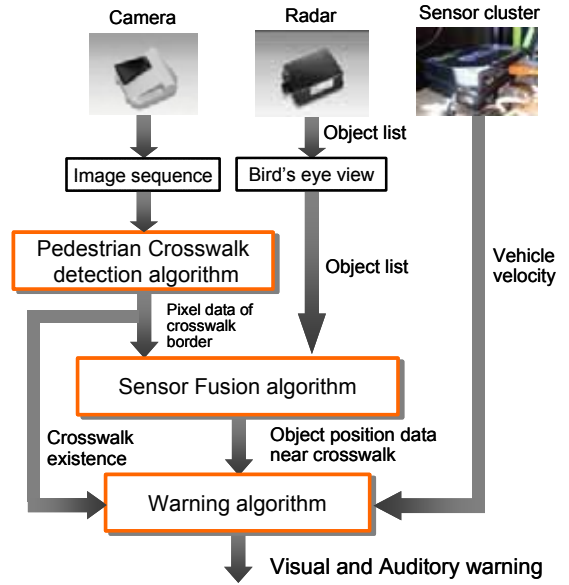


Fig.4 Flowchart of pedestrian detection and collision warning (ROC stands for “Region of Crosswalk”).

A. Crosswalk detection algorithm

For a crosswalk detection, a feature called “cross ratio” of four collinear points is utilized. Since the cross ratio is invariant under the perspective projection, the cross ratio of the crosswalk in 3D space is the same as that of its image, shown in Figure 5. Here, the cross ratio $[abcd]$ of four points a, b, c, d are calculated as follows: where the symbol written as $\overline{ab}$ refer to the distance between the point a and the point b .

$$[abcd] = \frac{\overline{ac}}{\overline{bc}} \cdot \frac{\overline{bd}}{\overline{ad}} \quad (1)$$

The flow of our crosswalk detection algorithm is shown in Figure 6. First, to extract boundaries of the white lines on a road, an image captured by the camera is processed by two algorithms: binarization and edge detection. To extract edges from the input image, the Canny’s algorithm is used. Next, the logical product of two outputs are calculated to reduce the errors due to shadow casted on the road. Then, vertical edges are classified into two kinds: the boundaries where the intensity changes from dark to bright or from bright to dark.

Finally, the cross ratio of the two pairs of boundaries are calculated. Because it is assumed that the car is going straight ahead, four points (a, b, c, d in Figure 5) which have the same vertical height in the image are extracted. In this step, the candidates of crosswalk in the image are reduced by using the knowledge of the periodicity of the crosswalk by checking the number of boundaries at the same height in the image and the width between two edges. Then, we calculate the cross ratios $[abcd]$ of four points a, b, c, d as indicated in Eq.(1). If three pairs edges extracted from the calculation results are adjusted to cross ratio of crosswalk. The extracted edges are regarded as a crosswalk.

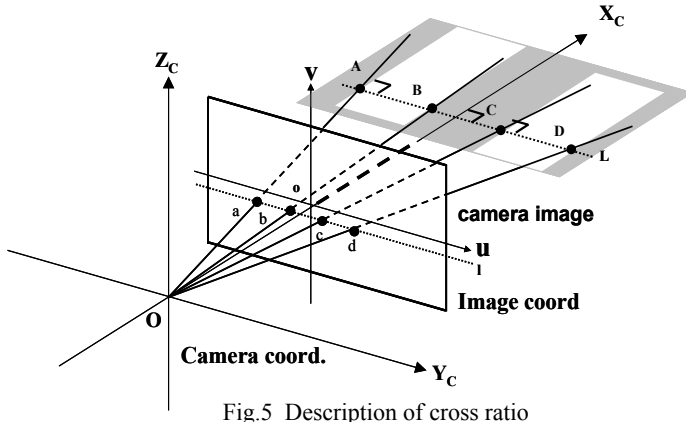


Fig.5 Description of cross ratio

B. Sensor fusion algorithm

The flow diagram of the sensor fusion algorithm and the example of the information processing by this algorithm are shown in Figure 7. The algorithm is based on the assumption that the vehicle is running straight and the vehicle direction is perpendicular to the crosswalk. The pixel coordinates of the boundary of the crosswalk detected from the image processing unit and the forward object position data with respect to the host vehicle obtained from the millimeter wave radar are used to detect moving objects near a crosswalk. The following procedures of the sensor data fusion algorithm are conducted for pedestrian detection task.

1) The pixel coordinates of the Region-of-Crosswalk boundary from the camera image are projected into the coordinate system of a bird's eye view. According to Figure 8, the relationship of the coordinates system of camera (u, v) and the coordinate system, the bird's eye view, of radar (X_r, Y_r, Z_r) can be expressed as follows:

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{pmatrix} \begin{pmatrix} X_r \\ Y_r \\ Z_r \\ 1 \end{pmatrix} \quad (2)$$

where $p_{11}, p_{12}, p_{13}, \dots, p_{34}$ are the elements of the perspective projection matrix for coordinate transformation. The matrix coefficients are determined based on the relationship between the position of a reference object information of the camera images and the measured position with the radar (calibration). In the calculation process, Z_r is set as a constant which corresponds to the sensor height with respect to the road plane height. In this paper, 78 different positions are used to determine the perspective projection matrix by Least Square Method.

2) The objects which are on or nearby a crosswalk are selected as candidates of a pedestrian. Here, the region of crosswalk is extended with a margin of 1 meter at each side in order to detect pedestrians nearby the crosswalk.

3) From the previous step, the radar data (X, Z) which

indicates candidates of pedestrians are projected into the coordinate system of the camera (u, v) , and the group of pixel corners (u_i, v_i) are connected as a rectangle. The written rectangle size is determined by width W and height H . The rectangular box becomes the Region-of-Interest (ROI). If an appearance-based pedestrian algorithm is available for advanced detection, the algorithm could be done in the candidate rectangular block regions instead of the whole region of an image to reduce the computational cost. However, this paper does not include that in the algorithm, as the moving objects data from the millimeter wave radar are used instead.

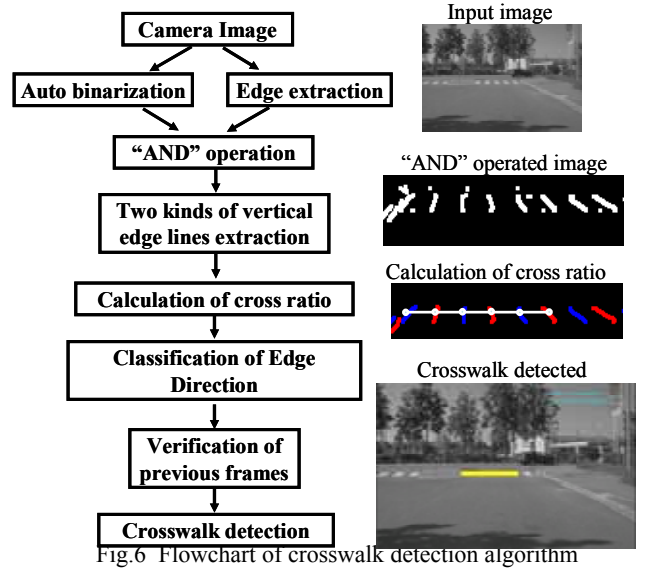


Fig.6 Flowchart of crosswalk detection algorithm

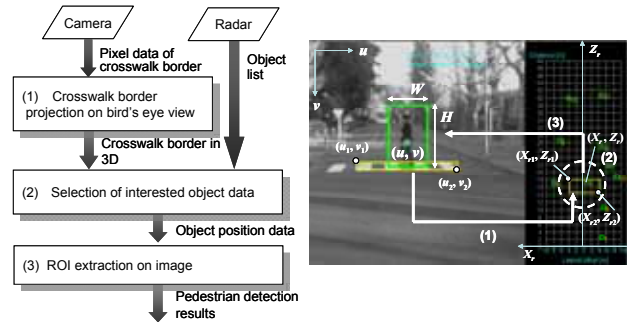


Fig.7 Flowchart of sensor fusion algorithm

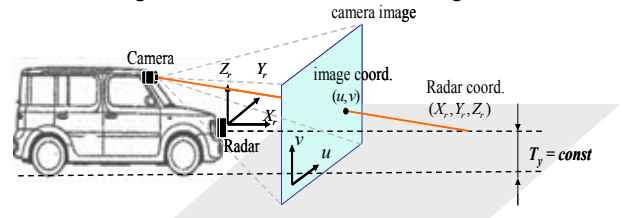


Fig.8 Camera and radar coordinate systems

C. Warning algorithm

This algorithm provides a visual and an auditory warning in real time to a driver depending on a collision risk to the pedestrian. The warning algorithm estimates the collision risk from the predicted position of the pedestrian, and the host vehicle velocity and the crosswalk information explained in the previous section. Conventional warning system proposed in [5] used a simple index such as Time-to-Collision (TTC) as a criterion whether the warning should be issued or not. However, TTC is the index which considers the relative speed between two vehicles in identical directions, so TTC may not be suitable to indicate the risk when the object approaches to the host vehicle in lateral direction. Until now, the systems based on the current pedestrian longitudinal position have been proposed to estimate the collision risk [6]. Therefore, the systems based on TTC index might give inappropriate warning to drivers since they do not take the lateral position of the pedestrian into account. Therefore, this research presents the collision risk estimation which combines TTC with the prediction of pedestrian position for critical warning to driver. The pictorial diagram is shown in Figure 9. First, the pedestrian motion direction is defined, and then the predicted time (T_p) for the host vehicle to reach this axis is calculated. Predicted pedestrian position is the position that a pedestrian moves for the time (T_p). The following assumptions are made for the calculation.

Pedestrians and the host vehicle will continue moving with constant velocity ($V_{ped} = const, V_{car} = const$).

Host vehicle is running straight and the vehicle direction is perpendicular to the crosswalk ($Y_{ca} = 0$).

The pedestrians move along the moving axis defined above with a constant average walking speed ($X_{ped} = const$).

The predicted time (T_p) according to these conditions can be expressed as follows:

$$T_p(t) = \frac{X_{ped}(t) - X_{car}(t)}{V_{car}} \quad (3)$$

And then the predicted pedestrian position can be calculated as follows:

$$Y_{ped}(t + T_p(t)) = Y_{ped}(t) + V_{ped}(t) \cdot T_p(t) \quad (4)$$

where, $X_{ped}(t) - X_{car}(t)$, $Y_{ped}(t)$ are position measured by millimeter wave radar. The pedestrian crossing speed (V_{ped}) is calculated by the differentiated value of the lateral position (Y_{ped}) and then the smoothing preprocessing for the differentiated signal are conducted by moving average filter. From the pedestrian current position and the vehicle speed acquired by sensors, the pedestrian predicted position can be calculated in real time.

This paper utilizes the position, the lateral velocity of the detected pedestrian in the design of the visual indication, in order to inform the driver about the presence of a crosswalk or a pedestrian. The interest region for detection is determined as shown in Figure 10. The top border of the interest region is set to be proportional to the vehicle speed T_{ic} indicates the parameter for setting the range of longitudinal top in the pedestrian detection which is

equivalent to time-to-collision, and V_{car} indicates the vehicle speed. The proportional relationship between the vehicle and longitudinal range T_{ic} is set to the constant value of 4.0 seconds. This value is based on Ueda's report which addresses that the warning of pedestrian protection algorithm is effective at time-to-collision of 4.0 seconds [7]. Visualized hazardous region is separated into the probability of pedestrian collision by every 1.0 seconds. Then, to check the data processing output of the system, a red/yellow/green light indicator in a vehicle-mounted display is built up to show the presence of crosswalk or a pedestrian as shown in Figure 11. Depending on the condition of the pedestrian position and the walking speed, the collision warning which consists of auditory warning, the color of the visual warning on the vehicle display is changed according to the following rules, and an auditory warning is issued to the driver when the situation becomes more critical as shown in Figure 12 and 13:

- If a crosswalk and pedestrians are not detected, the color is GREEN.
- If a crosswalk or static objects near the crosswalk is detected the color is YELLOW and the yellow color message "CROSSWALK DETECTED" is shown on the display.
- If a crosswalk and moving objects on the crosswalk are detected and the predicted position is under the value of $V_{car} \cdot T_{ic}$, the color is RED and the red color message "CROSSWALK OCCUPIED" is shown on the display, and if the value of T_{ic} is under 2.0 seconds, the auditory warning will be provided to the driver.

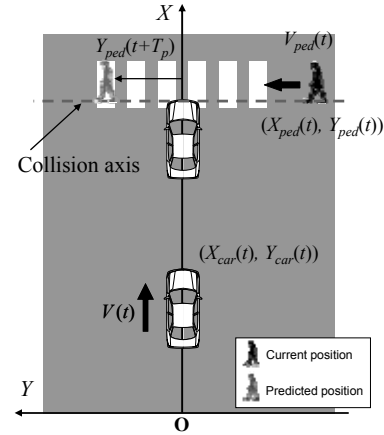


Fig.9 Description of predicted pedestrian protection

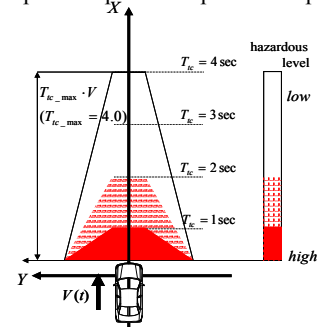


Fig.10 Definition of interest region and hazardous level

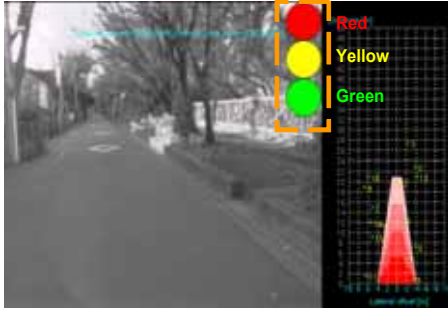


Fig.11 Information presentation on the display

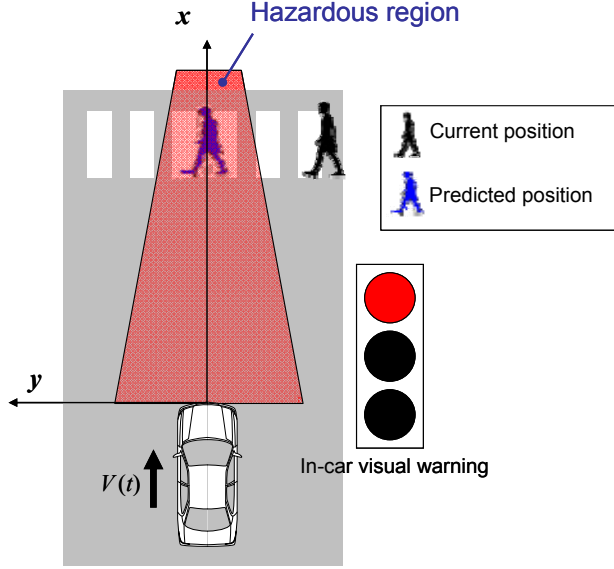


Fig.12 Pedestrian collision warning when the situation is critical

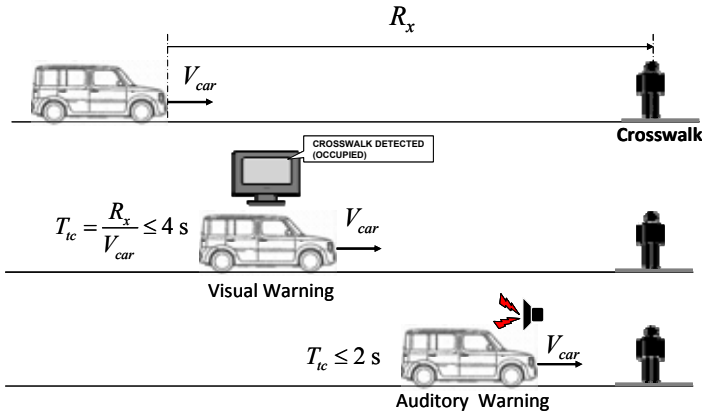


Fig.13 Description of pedestrian collision warning interfaces

IV. EXPERIMENTAL RESULTS

Experiments using the test vehicle were conducted to confirm whether the proposed pedestrian collision avoidance system can work in principle. The vehicle was running at the constant speed of 20 km/h in perpendicular to a crosswalk. As shown in Figure 14, the experimental condition was that a

pedestrian starts walking on the crosswalk if the time-to-collision T_{ic} is under 3.0 seconds and that the test vehicle does not slow down. We replaced a real pedestrian with puppet to secure the safety of the experiment. The experimental result is indicated in Figure 15 and 16. When the pedestrian entered the region of interest, the artificial traffic light on the display was changed from yellow to red. The trajectory of puppet with respect to the host vehicle position during the experiment is shown in Figure 17. According to Figure 17, the trajectory of puppet entered the region of interest (warning region). Therefore, the validity of the proposed pedestrian detection and warning system algorithm was verified.

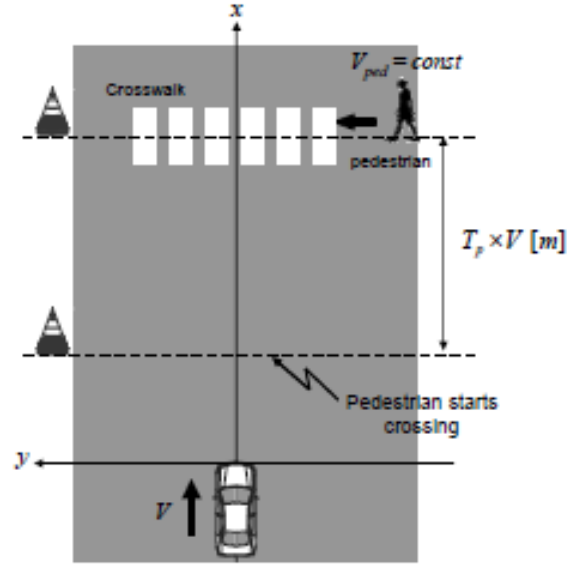


Fig.14 Experimental condition

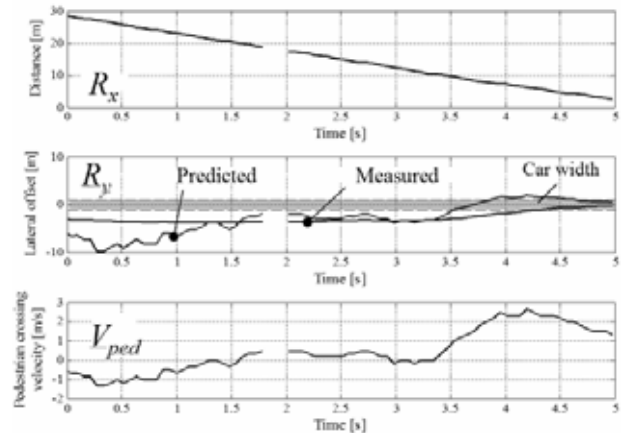


Fig.15 Experimental result of pedestrian detection



Fig.16 Snapshots of pedestrian collision warning experiment

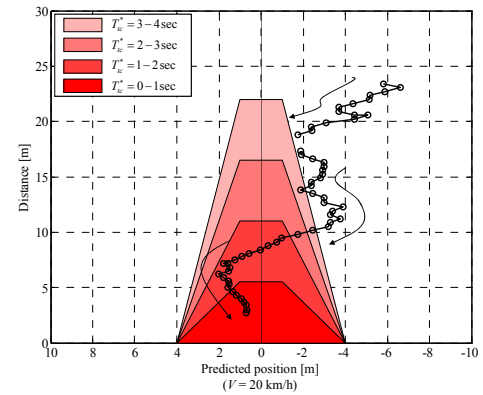


Fig.17 Transition of predicted pedestrian position (bird's eye view)

V. CONCLUSIONS

This paper described an algorithm for implementing a pedestrian detection and collision warning system for an experimental vehicle, limited to locations nearby crosswalks. The described system is promising since it is also easy to understand for the driver. The design principle used here to handle the pedestrian detection problem is based on the assumption that objects moving along a crosswalk can be inferred as pedestrians under certain circumstances. As a method for pedestrian collision prevention, visual and auditory warning systems are issued to drivers in real time based on the pedestrian position and movement, with respect to the host vehicle motion. The sensor setup on the experimental vehicle for pedestrian detection was presented and experimental results reveal that the proposed sensor fusion approach for pedestrian collision warning system can work in practice.

Further investigations in public road driving will be conducted by using the same experimental setup indicated in this paper, and the evaluation of false alarm rates and missing detections in real-world driving will be conducted.

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